-Project Deliverables - Grading Distribution –Submission Guidelines – Project Topics - Deadlines-

CMPS 380 Cybersecurity Fundamentals
Fall 2026

Two Project Deliverables (Blackboard Submission Deadline: Midnight 14-11-2026):

1. A Project Report (MS-Word file), and
2. A Video Recording, showing the implementation/ prototype/application/demo in detail. (See next section for details)

Various **project topics** are listed at the end of this document

Details materials to be submitted

Only one member of each group MUST upload **only one copy** of the following deliverables on Blackboard in two separate subfolders (*Project Report,* and *Video*) created by the instructor:

1. **A Project Report *in MS-Word* (not PDF)**: It must include the following items in a single Word document.
   - **a.** Step by step documentation of the project with the project objective, application use case, what and how you did the project including all commands for setting up the environment and the implementation/application/prototype (Proof of Concepts) details, technology/tools used, limitation of your project, impact in the society of your project, and what other people will learn from your project (*see the grading distribution table in the next page for details*); This part of the report should not exceed **3,500 words** without Appendix.
   - **b.** **Appendix**: **Student Reflection and User Manual**: In this appendix, (a) each group member will summarize his/her specific contributions to and tasks of this project, problems faced if any, and what he/she has learned from this project, etc.; (b) the group will provide a user manual on how to use the implementation/prototype/application/demo (Proof of Concepts). Also include any source code created/written or used from other resources in this Appendix. Provide also commands on how to use various functions/use cases of the demo(Proof of Concepts).

   ** The Appendix has no word limit. The Appendix must be submitted with the Report as a single Word file.

- Provide a list of references at the end of your Report and use the reference number in the body of the text in the report including in the Appendix. An example of the reference list is given below.
- References

[1] G. Eason, B. Noble, and I.N. Sneddon, "On certain integrals of Lipschitz-Hankel type involving products of Bessel functions," Phil. Trans. Roy. Soc. London, vol. A247, pp. 529-551, April 1955. (*references*)

[2] I.S. Jacobs and C.P. Bean, "Fine particles, thin films and exchange anisotropy," in Magnetism, vol. III, G.T. Rado and H. Suhl, Eds. New York: Academic, 1963, pp. 271-350

How the reference numbers can be used in the body of the report is shown in the following example:

- We have used Bessel Functions [1] to do this project. It is mentioned in [2] that theses are very effective.

2. **A Video** recorded before the actual live presentation (demo) in the class (less than 12 minutes running time). The video will capture the demonstration of the project such as prototype/implementation/demo (Proof of Concepts) with appropriate application/use case(s)/scenario(s) of the chosen topic of the group. Make sure that your video has your background narrative (running commentary). You may distort your voice if you want to make your voice anonymous. A zero grade will be assigned if your video has no running background narrative.

   All demonstrations/implementation (Proof of Concepts) should be demonstrated on any operating environment. Students may also choose any Linux Distribution of their choice, e.g., Kali, Parrot OS, Fedora, Ubuntu, Debian, CentOS, Manjaro, CSI Linux, etc. For a virtual machine, students can use any free solution, e.g., VirtualBox, VMware Workstation Player. UTM is recommended, if you are using an Appel Mac Book Pro with ARM Processors M1/M2.

- The Cover Page of your Report must include the following information. 2.0 grade will be deducted for each of the following information is missing from your submissions.
  a) Group ID, All students' names, their QUID.
  b) The Total Word count of the report *excluding* Appendix.
  c) Contribution percentage of each group member such as an example below:
  `Ahmed (20%); Mohamed (20%); Rashid (20%); Amr (20%); Salem (20%),` if all members contributed roughly equal. Otherwise, the percentage will vary. See page 3 for details.
  d) The video must also have the Group ID, names and QUID of each member in the opening frame.
  e) The video must be in good quality, all information and texts are easily visible, readable and include background narrative.

The Preparation of the Report and the Submission Requirements are given latter in this document.

Important Dates

- All Components (The Report and the Video) Submission Deadline: **14-11-2026.**
- Presentations: **Weeks 12 & 13** during your class time.
- All students must attend all presentations delivered by other groups, otherwise an absence will be counted for each missed class.

Assessment

Any evidence of plagiarisms, such as copied parts or whole from other project groups (past or current year), other sources, or missing references to any external resources if used, their work in this project will lead to **0 point to Project** and probably removal from the course!

Grading Distribution:

Overall Report structure, clarity, margin, fonts and documentation quality (2%)	**Report (10%)**
Req. #1: Project description (2%) - Introduction, project objectives, and key applications/use cases/scenarios, references.	
Req. #2: Explain Implementation: (2%) - Implementation/prototype/demo of the use case(s). - Details about the environment, program, tool(s) and technology used - Learning outcomes - Novelty, and quality.	
Req. #3: (2%) - Overall Impact (usefulness, harmful etc.) of your project on computer security, society at large, on the general public, and what they can learn, limitation of your project, etc.	
Appendix (2%) Req. #4: Reflection. The contribution of each individual member. Req. #5: Code/user manual/commands/technologies/ environment set up, etc.	
Video and Live Demonstration: - **Video** (4%): - Recording quality, background narrative, understandability, group information, logical sequence of use cases, clear description of the tools used, reasons for using those tools, and overall clarity. - **Live demonstration** (6%): (no PowerPoint slides) - Implemented/demo/prototyped, quality explanation - Clear, conciseness, understandability, tools selection, reasons for using those tools, Q&A, confidence in the presentation - **Project complexity** (5%). - More complex project and use cases which require more efforts, varieties of tools used, good application/scenario and creativity used attract better grade. See Grading rubrics on page 3.	Video and Live Demonstration of the Proof of Concepts (15%)
Total:	**25%**

Group Formation and Presentation

- Each group must have three to four members. The students are required to find their group.
- Once you have formed your group, select your project topic from the topic-list given on Page 4. Then do the following: (a) *One Group member will send me the following by email by* ***12-09-2026****,* and (b) *Cc the email to other Group members*:
  - The details of each Group member (name, email address, and QUID), and
  - The project topic. You can change the topic later.
- Each group will have total 20 mins to demonstrate their project and answer questions.
  - 12 minutes – Live demonstration of prototype/application/use cases/implementation.
  - 8 minutes – Questions and Answers (Q&A).
- All students must attend all presentations delivered by other groups, otherwise an absence will be counted for each missed class.

Grading rubrics

Grade	**All grades submitted on Blackboard are Final. Every project will receive a written assessment from the instructor. No request for grade review or reassessment will be entertained.**
90 - 100%	**Excellent** Report with very good logical flow of sections, to-the-point details information, factual information, a good number of references used, clear implementation details, *very complex* topic, scenarios/use cases and using tools and techniques, outstanding video, outstanding presentation, excellent demonstration with novelty, originality, high complexity of the topic/demo, varieties of relevant tools used, and the expected effort of the group.
80 - 89%	**Very good** report and video but with some weak logical explanation or justification, provided information is not quite complete, the factual information could include some more information. Sometimes the description may not be clear. The topic, scenarios/use cases and using tools and techniques are *moderately complex*, video is reasonable, the presentation is ok, but not excellent.
70 - 79%	**Good** but with several incomplete information and weak description. The explanation may not be very logical and/or realistic. The topic, scenarios/use cases and using tools and techniques are *not complex*, the video is not very good, the presentation has problems in terms of contents and logical flow.
0 - 69%	**Not satisfactory** with a lack of clear information, many facts are missing, presentation is not good, practical aspect is missing or incomplete, the quality of the video is not acceptable, the topic, scenarios/use cases and using tools and techniques are *not complex at all*, and too obvious. Vital information is missing, the presentation was no logical and artificial.

** "*Very complex*" means it requires creativity, a considerable amount of time and effort to understand, learn and use.

Report preparation: Structure your report into sections and subsections such as Title, Introduction/motivation/objective, application/use cases/scenarios, information on the implementation/prototype, description, impact, learning outcomes, references, etc. Check the grading distribution table (page 2) and the grading rubrics (page 3). Provide example(s), tables and diagrams if needed. Use either *Times New Roman* or *Arial* font style with 11 font size. Your report should use a single-column format with 1.0 line spacing. Provide references at the end of your main Report.

Submission Requirements: Your group/team MUST comply with the following submission requirements; **otherwise, grade will be deducted**:

1. **Ensure that only one group member submits one *electronic copy*** of the report and video.
2. Your report, including the appendix, must not have more than 20% similarity to other sources. If it does, the grade will be deducted significantly.
3. The cover page of your Report MUST include a peer group evaluation with an estimate of each member's contribution to the project. Write the percentage contribution for each group member, so it adds up to 100%. You *may* use this evaluation to adjust the grades awarded to each group member. All group members must agree on the given effort values. If any team member disagrees, the group will email me their disagreement directly.

• Effort distribution of the students, as shown below, as an example:

QUID:_____STUDENT NAME: _____ _____Effort given___%
QUID:_____STUDENT NAME: _____ Effort given___%
QUID:_____STUDENT NAME: _____ Effort given___%
QUID:_____STUDENT NAME: _____ Effort given___%

Suggested Project Topics

The following are some suggested topics; you may select another topic for your project that is not included here. If so, please email me your topic so I can review it and determine whether it is suitable for this course. You must demonstrate proof of concept, the program if there is any, and the list of tools that you used in your project. You are responsible for learning the tools associated with your project topic. Many resources and examples are available from YouTube, Cryptool.org, Kali Linux, TryHackMe, Python libraries, CSI Linux, and other sources. YouTube is the best option to learn. Once you have selected your project topic from the list below, do the following: (a) *One Group member will send me the following by email by* ***12-09-2026****,* and (b) *Cc the email to other Group members*:

- The details of each Group member (name – email address- QUID), and
- The project topic. You can change the topic later.

Topics	Task Description
Vulnerabilities and/or Exploits (from 2020 to 2024, refer to CVEs https://cve.mitre.org/)	Introduce and describe one selected vulnerability and/or exploit. The description should be based on the CVSS, https://www.first.org/cvss/. Demonstrate in a Proof of Concept how the vulnerability and/or exploit works by using an adequate use case.
Penetration Testing Tools, e.g., Tools available from Kali Linux, TryHackMe, or Parrot OS	Introduce and describe one selected offensive tool. There are many available in the platform: https://www.kali.org. https://tryhackme.com/ is another platform for offensive exercises and learning tool. Demonstrate in a Proof of Concept an attack or one or more penetration testing features by using the tool in an adequate use case. Examples of tools are nmap, wireshark, password cracking tools such as HashCat, THC Hydra, John the Ripper, Nikto, Brutus, Medusa, Nessus, Aircrack-ng, Metasploit framework, etc.
Cryptographic Algorithms	Introduce and describe symmetric or asymmetric cryptographic algorithms, e.g., for encryption/decryption, for signing/verifying or for hashing. The description should emphasize the mathematical difficult problem that the algorithm is based on. You can also choose to compare various algorithms of encryption/decryption in terms of their speed and efficiency. You can use the codes/programs that are available from various sources. Demonstrate in a Proof of Concept how the algorithm works by using an adequate use case. Tools, such as https://docs.python.org/3/library/crypto.html https://pypi.org/project/cryptography/ , https://cryptography.io/en/latest/ https://www.cryptool.org/en/ These can be used for the purpose of demonstration.
Authentication and Authorization	Introduce and describe one biometric authentication and authorization method, e.g., Fingerprint, Face Authentication, Iris, Voice, or a combination of them. Demonstrate in a Proof of Concept how the protocol works by using an adequate use case.
Access Control	Introduce and describe an access control method implemented on Linux, either Role-Based Access Control (RBAC) or Attribute-Based Access Control (ABAC). The description should cover all features, and components.

	Demonstrate in a Proof of Concept how the method (tool) works by using an adequate use case.
Malware and Ransomware	Choose from below one type of malware, and demonstrate the infection and effects of the malware on a virtual machine, together with detection methods/tools. - Ransomware - Bots/botnets - Spyware/adware - Rootkit - Virus - Worm - Trojan, etc.

Be aware of

- You must refrain from copying the project from other groups or from old reports done by other students or other electronic or paper sources. Copying part or all of it from other sources or groups may result in withdrawal from the course with a 0 grade.
- The report must be clearly and easily readable. Otherwise, you will receive a 0 grade for unreadable or unclear text.
- If you use any AI tools, you MUST include this as a reference and acknowledge this explicitly in your report.

University Code of Conduct

QU expects its students to adopt and abide by the highest standards of conduct in their interaction with professors, peers, staff members and the wider university community. Moreover, QU expects its students to act maturely and responsibly in their relationships with others. Every student is expected to assume the obligations and responsibilities required from them for being members of the QU community.

As such, a student is expected not to engage in behaviors that compromise their integrity, as well as the integrity of QU. Further information regarding the University Code of Conduct may be found on the web at http://www.qu.edu.qa/students/code-of-conduct

Disclaimer: The above may change. If so, you will be notified.

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